

List of Design Patterns

A Complete Reference Catalogue

GoF · SOLID/OO · Enterprise · Integration · Microservices · Distributed Systems
Event-Driven · Cloud-Native · Caching · API · Database · Concurrency
Architecture · Functional · UI · Testing · Security · Kafka · Kubernetes · Workflow

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1. GoF — Gang of Four: 23 classic patterns

CREATIONAL — 5

- Abstract Factory
- Builder
- Factory Method
- Prototype
- Singleton

STRUCTURAL — 7

- Adapter
- Bridge
- Composite
- Decorator
- Facade
- Flyweight
- Proxy

BEHAVIORAL — 11

- Chain of Responsibility
- Command
- Interpreter
- Iterator
- Mediator
- Memento
- Observer
- State
- Strategy
- Template Method
- Visitor

These are the 23 patterns from Design Patterns: Elements of Reusable Object-Oriented Software.

2. SOLID-related / Object-Oriented Design Patterns

These aren't one official catalog like GoF, but commonly discussed OO techniques/patterns include:

- Dependency Injection
- Dependency Inversion
- Composition over Inheritance
- Null Object
- Object Pool
- Lazy Initialization
- Multiton
- Registry
- Service Locator
- Special Case
- Extension Object
- Role Object
- Marker Interface
- Private Class Data
- Delegation
- Subclass Sandbox

3. Enterprise Application Patterns — Martin Fowler

From Patterns of Enterprise Application Architecture.

DOMAIN LOGIC

- Transaction Script
- Domain Model
- Table Module
- Service Layer

DATA SOURCE ARCHITECTURAL PATTERNS

- Table Data Gateway
- Row Data Gateway
- Active Record
- Data Mapper

OBJECT-RELATIONAL BEHAVIORAL PATTERNS

- Unit of Work
- Identity Map
- Lazy Load

OBJECT-RELATIONAL STRUCTURAL PATTERNS

- Identity Field
- Foreign Key Mapping
- Association Table Mapping
- Dependent Mapping
- Embedded Value
- Serialized LOB
- Single Table Inheritance
- Class Table Inheritance
- Concrete Table Inheritance

WEB PRESENTATION

- Model View Controller
- Page Controller
- Front Controller
- Template View
- Transform View
- Two-Step View
- Application Controller

DISTRIBUTION

- Remote Facade
- Data Transfer Object
- Gateway
- Service Stub

OFFLINE CONCURRENCY

- Optimistic Offline Lock
- Pessimistic Offline Lock
- Coarse-Grained Lock
- Implicit Lock

SESSION STATE

- Client Session State
- Server Session State
- Database Session State

BASE PATTERNS

- Gateway
- Mapper
- Layer Supertype
- Separated Interface
- Registry
- Value Object
- Money
- Special Case
- Plugin
- Service Stub

4. Enterprise Integration Patterns — Gregor Hohpe

Very important for Spring Boot + Kafka work.

MESSAGING CHANNELS

- Point-to-Point Channel
- Publish-Subscribe Channel
- Datatype Channel
- Invalid Message Channel
- Dead Letter Channel
- Guaranteed Delivery
- Message Bridge
- Message Bus

MESSAGE CONSTRUCTION

- Command Message
- Document Message
- Event Message
- Request-Reply
- Return Address
- Correlation Identifier
- Message Sequence
- Message Expiration
- Format Indicator

MESSAGE ROUTING

- Content-Based Router
- Message Filter
- Recipient List
- Splitter
- Aggregator
- Resequencer
- Composed Message Processor
- Scatter-Gather
- Routing Slip
- Process Manager
- Dynamic Router

MESSAGE TRANSFORMATION

- Message Translator
- Normalizer
- Canonical Data Model
- Envelope Wrapper
- Claim Check

MESSAGING ENDPOINTS

- Messaging Gateway
- Messaging Mapper
- Transactional Client

- Polling Consumer
- Event-Driven Consumer
- Competing Consumers
- Message Dispatcher
- Selective Consumer
- Durable Subscriber
- Idempotent Receiver
- Service Activator

SYSTEM MANAGEMENT

- Wire Tap
- Control Bus
- Detour
- Message History
- Message Store
- Smart Proxy
- Test Message
- Channel Purger

5. Microservices Patterns

DECOMPOSITION

- Decompose by Business Capability
- Decompose by Subdomain
- Strangler Fig
- Parallel Run
- Anti-Corruption Layer

COMMUNICATION

- Remote Procedure Invocation
- Messaging
- API Gateway
- Backend for Frontend
- Service Mesh

DATA

- Database per Service
- Shared Database
- Saga
- Transactional Outbox
- Polling Publisher
- Change Data Capture
- Event Sourcing
- CQRS
- Materialized View

RELIABILITY

- Circuit Breaker
- Retry
- Timeout
- Bulkhead
- Rate Limiter
- Fallback
- Health Check
- Service Discovery
- Self Registration
- Third-Party Registration

DEPLOYMENT

- Multiple Service Instances per Host
- Single Service Instance per Host
- Service Instance per VM
- Service Instance per Container
- Serverless Deployment
- Sidecar
- Ambassador
- Init Container

6. Distributed Systems Patterns

- Leader Election
- Distributed Lock
- Lease
- Heartbeat
- Quorum
- Read Repair
- Anti-Entropy
- Gossip
- Consistent Hashing
- Sharding
- Replication
- Primary-Replica
- Multi-Leader Replication
- Leaderless Replication
- Failover
- Retry
- Exponential Backoff
- Jitter
- Idempotency
- Deduplication
- Exactly-Once Processing
- At-Least-Once Processing
- At-Most-Once Processing

7. Event-Driven Architecture Patterns

- Event Notification
- Event-Carried State Transfer
- Event Sourcing
- CQRS
- Event Streaming
- Event Choreography
- Event Orchestration
- Saga
- Transactional Outbox
- Inbox Pattern
- Dead Letter Queue
- Retry Queue
- Delayed Message
- Poison Message Handling
- Idempotent Consumer
- Event Replay
- Snapshotting
- Event Versioning
- Schema Evolution

8. Cloud / Cloud-Native Patterns

- Sidecar
- Ambassador
- Adapter
- Anti-Corruption Layer
- External Configuration Store
- Service Discovery
- Gateway Routing
- Gateway Aggregation
- Gateway Offloading
- Health Endpoint Monitoring
- Retry
- Circuit Breaker
- Bulkhead
- Throttling
- Queue-Based Load Leveling
- Competing Consumers
- Cache-Aside
- CQRS
- Event Sourcing
- Valet Key
- Priority Queue
- Scheduler Agent Supervisor

9. Caching Patterns

- Cache-Aside
- Read-Through Cache
- Write-Through Cache
- Write-Behind / Write-Back
- Refresh-Ahead
- Cache-Aside with TTL
- Distributed Cache
- Near Cache
- Cache Stampede Prevention
- Cache Warming
- Cache Invalidation
- Negative Caching

10. API Patterns

- API Gateway
- Backend for Frontend
- Aggregator
- Adapter
- Facade
- Strangler
- Versioned API
- Pagination
- Filtering
- Sorting
- Cursor Pagination
- HATEOAS
- Idempotency Key
- Request Validation
- Rate Limiting
- Request Hedging
- Bulk API
- Partial Response
- Long-Running Operation
- Polling
- Webhook / Callback

11. Database Patterns

- Repository
- DAO
- Active Record
- Data Mapper
- Unit of Work
- Identity Map
- Specification
- Query Object
- CQRS

- Event Sourcing
- Database per Service
- Shared Database
- Read Replica
- Sharding
- Partitioning
- Materialized View
- Transactional Outbox
- Inbox
- CDC — Change Data Capture

12. Concurrency Patterns

- Immutable Object
- Monitor Object
- Active Object
- Half-Sync/Half-Async
- Leader/Followers
- Thread Pool
- Producer-Consumer
- Read-Write Lock
- Double-Checked Locking
- Guarded Suspension
- Balking
- Scheduler
- Future
- Promise
- Actor Model
- Reactor
- Proactor

13. Architectural Patterns

These are bigger than normal class-level design patterns.

- Layered Architecture
- N-Tier Architecture
- Hexagonal Architecture
- Clean Architecture
- Onion Architecture
- Ports and Adapters
- MVC
- MVP
- MVVM
- Microkernel / Plugin Architecture
- Event-Driven Architecture
- Microservices Architecture
- Monolithic Architecture
- Modular Monolith
- Service-Oriented Architecture
- Pipe-and-Filter
- Broker Architecture
- Peer-to-Peer
- Space-Based Architecture
- Client-Server
- Master-Slave / Primary-Replica
- CQRS Architecture
- Serverless Architecture
- Cell-Based Architecture

14. Functional Programming Patterns

- Function Composition
- Higher-Order Function
- Currying
- Partial Application
- Memoization
- Monad
- Functor
- Applicative
- Either
- Maybe / Optional
- Reader
- Writer
- State
- Fold / Reduce
- Map
- Filter
- Pipeline

15. UI / Frontend Patterns

- Container-Presentational
- Compound Components
- Render Props
- Higher-Order Component
- Provider Pattern
- State Reducer
- Controlled Component
- Uncontrolled Component
- Flux
- Observer
- Pub/Sub
- Model-View-ViewModel
- Module Pattern
- Revealing Module
- Facade
- Adapter

16. Testing Patterns

- Test Double
- Dummy
- Stub
- Fake
- Mock
- Spy
- Test Fixture
- Object Mother
- Test Data Builder
- Given-When-Then
- Arrange-Act-Assert
- Test Pyramid
- Contract Testing
- Consumer-Driven Contract
- Snapshot Testing

17. Security Patterns

- Authentication Gateway
- Authorization Gateway
- Access Control List
- Role-Based Access Control
- Attribute-Based Access Control
- Token-Based Authentication
- API Key
- Capability
- Secure Proxy
- Input Validation
- Secure Session
- Password Hashing
- Secrets Vault
- Zero Trust
- Defense in Depth

18. Messaging / Kafka-specific Patterns

Worth knowing separately when learning Kafka deeply.

- Consumer Group
- Competing Consumers
- Partitioned Consumer
- Publish-Subscribe
- Event Streaming
- Event Sourcing
- Transactional Outbox
- Inbox
- Idempotent Consumer
- Dead Letter Topic
- Retry Topic
- Poison Pill Handling
- Message Keying
- Partitioning
- Consumer Rebalancing
- Exactly-Once Semantics
- At-Least-Once Delivery
- Event Replay
- Schema Evolution
- Schema Registry
- Stream Processing
- Stream/Table Duality
- Kappa Architecture

19. Kubernetes / Infrastructure Patterns

- Sidecar
- Ambassador
- Adapter
- Init Container
- Controller Pattern
- Operator Pattern
- Reconciliation Loop
- Desired State
- Declarative Configuration
- Service Discovery
- Health Check
- Readiness
- Liveness
- Startup Probe
- Rolling Deployment
- Blue-Green Deployment
- Canary Deployment
- A/B Deployment
- DaemonSet Pattern
- Job Pattern
- CronJob Pattern
- Pod-per-Service
- Sidecar Proxy
- Service Mesh

20. Integration / Workflow Patterns

- Saga
- Process Manager
- Orchestrator
- Choreography
- Compensating Transaction
- Retry

- Timeout
- Circuit Breaker
- Bulkhead
- Fallback
- Idempotency

- Correlation ID
- Distributed Transaction
- Two-Phase Commit
- Transactional Messaging
- Outbox

- Inbox
- Workflow Engine
- Long-Running Process